

Hallar todos los $x \in \mathbb{R}$ tales que

PIN WASH
IAM 1P 2C N1

$$|2x - 10| \geq x^2 - 7x + 10$$

$$|2x - 10| = \begin{cases} 2x - 10 & \text{si } 2x - 10 \geq 0 \rightarrow x \geq 5 \text{ (Caso 1)} \\ -(2x - 10) & \text{si } 2x - 10 < 0 \rightarrow x < 5 \text{ (Caso 2)} \end{cases}$$

Caso 1

$$2x - 10 \geq x^2 - 7x + 10$$

$$0 \geq x^2 - 7x + 10 + 10 - 2x$$

$$0 \geq x^2 - 9x + 20$$

$$|2x - 10| = 2x - 10 \text{ si } 2x - 10 \geq 0 \rightarrow x \geq 5 \rightarrow \boxed{5 \geq 5}$$

Caso 2

$$-(2x - 10) \geq x^2 - 7x + 10$$

$$-2x + 10 \geq x^2 - 7x + 10$$

$$0 \geq x^2 - 7x + 10 + 2x - 10$$

$$0 \geq x^2 - 5x$$

$$|2x - 10| = -(2x - 10) \text{ si } 2x - 10 < 0 \rightarrow x < 5 \rightarrow \boxed{0 < 5}$$

$$|2x - 10| \geq x^2 - 7x + 10 \text{ si } x \in [0; 5]$$

SOLUCIÓN FINAL: $x \in [0; 5]$

C. A

$$\frac{-(-9) \pm \sqrt{(-9)^2 - 4 \cdot 1 \cdot (20)}}{2 \cdot 1} =$$

$$\frac{9 \pm \sqrt{81 - 80}}{2} =$$

$$\frac{9 \pm \sqrt{1}}{2} =$$

$$\frac{9 \pm 1}{2} \rightarrow \begin{matrix} 5 \\ 4 \end{matrix}$$

$$\frac{-(-5) \pm \sqrt{(-5)^2 - 4 \cdot 1 \cdot 0}}{2 \cdot 1} =$$

$$\frac{5 \pm \sqrt{25}}{2} =$$

$$\frac{5 \pm 5}{2} \rightarrow \begin{matrix} 5 \\ 0 \end{matrix}$$